

# 12.683

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**Question:**

The unit normal vector to the surface  $X^2 + Y^2 + Z^2 - 48 = 0$  at the point  $(4, 4, 4)$  is \_\_\_\_\_.

**Solution:**

Let the surface be defined by a function  $f(\mathbf{x}) = 0$ , where

$$\mathbf{x} = \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$$

$$f(\mathbf{x}) = \mathbf{x}^\top \mathbf{x} - 48 \quad (1)$$

The normal vector to the surface is given by the gradient of the function,  $\nabla f$ .

$$\nabla f = 2\mathbf{x} = \begin{pmatrix} 2X \\ 2Y \\ 2Z \end{pmatrix} \quad (2)$$

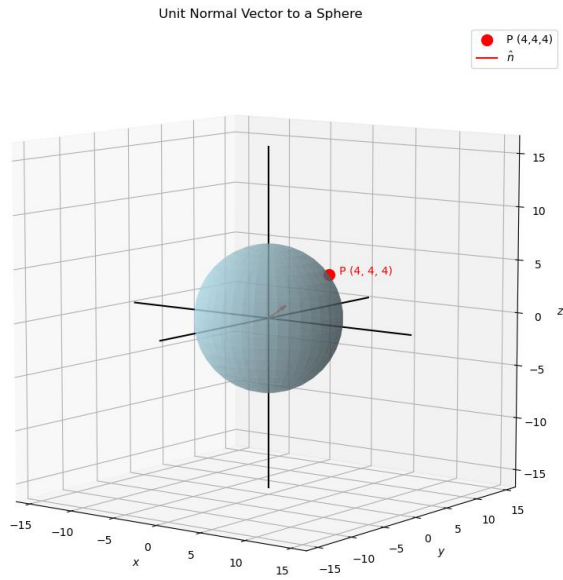
Substituting the point  $\mathbf{p} = \begin{pmatrix} 4 \\ 4 \\ 4 \end{pmatrix}$  into (2):

$$\mathbf{n} = \nabla f(\mathbf{p}) = 2\mathbf{p} = 2 \begin{pmatrix} 4 \\ 4 \\ 4 \end{pmatrix} = \begin{pmatrix} 8 \\ 8 \\ 8 \end{pmatrix} \quad (3)$$

To find the unit normal vector  $\hat{\mathbf{n}}$ ,

$$\|\mathbf{n}\| = \left\| \begin{pmatrix} 8 \\ 8 \\ 8 \end{pmatrix} \right\| = \sqrt{8^2 + 8^2 + 8^2} = \sqrt{3 \times 64} = 8\sqrt{3} \quad (4)$$

$$\hat{\mathbf{n}} = \frac{\mathbf{n}}{\|\mathbf{n}\|} = \frac{1}{8\sqrt{3}} \begin{pmatrix} 8 \\ 8 \\ 8 \end{pmatrix} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad (5)$$



Plot